GOOD AND BAD REDUCTION: EXAMPLES, WITH SOLUTIONS

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An elliptic curve over $\mathbb{Q}_p$ has good, multiplicative or additive reduction, and multiplicative reduction is split or nonsplit [Sil09, §VII.5]. The problems below run through concrete curves of each kind: Problem 1.8 reads the type off an equation, Problem 2.2 counts the rational points of the singular reduction, and Problem 3.1 changes the type by twisting the curve or extending the field. Problem 4.1 ($\star$) goes further: the curves Frey attached to solutions of $A + B = C$ have no additive reduction at all. Each problem is preceded by the definitions and results it uses.

Standing notation.

- A prime p, with $v = v_p$ its valuation on $\mathbb{Q}_p$, extended to $\overline{\mathbb{Q}}_p$.
- An elliptic curve E over $\mathbb{Q}_p$, given by a Weierstrass equation with coefficients in $\mathbb{Z}_p$, with discriminant Δ and invariants c_4, c_6 [Sil09, §III.1].
- When the equation is minimal, E_p is its reduction modulo p, a Weierstrass cubic over $\mathbb{F}_p$.
- We write
$$a_p \coloneqq p + 1 - \#E_p(\mathbb{F}_p),$$
counting the singular point too.

How to use this. Allow about 15 minutes for each of Problems 1.8, 2.2 and 3.1. Problem 4.1 is for home. The one `Magma` block, in Problem 1.8, runs in seconds in the free online calculator. Everything else is by hand.

1. Reading off the type

The following definition and results hold for every prime p, and are used throughout. In them, K is $\mathbb{Q}_p$ or a finite extension of it, with normalized valuation v_K, ring of integers $\mathcal{O}_K$ and residue field k, and E is an elliptic curve over K given by a Weierstrass equation with coefficients in $\mathcal{O}_K$.

Definition 1.1 (Reduction types [Sil09, §VII.5])**.** Suppose that the equation is minimal, and consider its reduction, a Weierstrass cubic over k. Then E has good reduction if the reduction is nonsingular, multiplicative reduction if the reduction has a node, and additive reduction if it has a cusp. Multiplicative reduction is split if the slopes of the tangent lines at the node lie in k, and nonsplit otherwise.

Proposition 1.2 (Discriminant and c_4 [Sil09, §III.1])**.** *For an equation $y^2 = x^3 + a_2x^2 + a_4x + a_6$ (that is, $a_1 = a_3 = 0$),*

$$\Delta = -16\big(4a_2^3a_6 - a_2^2a_4^2 - 18a_2a_4a_6 + 4a_4^3 + 27a_6^2\big), \qquad c_4 = 16\big(a_2^2 - 3a_4\big), \tag{1.1}$$

which follow from the formulas of.

Proposition 1.3 (Minimal equations [Sil09, Remark VII.1.1])**.** *Any other Weierstrass equation for E has discriminant $u^{-12}\Delta$ and invariant $u^{-4}c_4$ for some $u \in K^\times$. So if $v_K(\Delta) < 12$ or $v_K(c_4) < 4$, the equation is minimal.*

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Proposition 1.4 (Singular Weierstrass cubics [Sil09, Proposition III.1.4(a)]). *A Weierstrass cubic over a field with $\Delta = 0$ has exactly one singular point over an algebraic closure. It is a node, with two distinct tangent lines, if $c_4 \neq 0$, and a cusp, with one tangent line, if $c_4 = 0$.*

Proposition 1.5 (Criteria for the reduction type [Sil09, Proposition VII.5.1]). *Suppose that the equation is minimal. Then E has good reduction if $v_K(\Delta) = 0$, multiplicative reduction if $v_K(\Delta) > 0$ and $v_K(c_4) = 0$, and additive reduction if $v_K(\Delta) > 0$ and $v_K(c_4) > 0$.*

Proposition 1.6 (Extending the field [Sil09, Proposition VII.5.4(b)]). *Let L be a finite extension of K with ramification index e, so that $v_L = e\, v_K$ on K. If E has good, respectively multiplicative, reduction, then so does E_L.*

Proposition 1.7 (Potential good reduction [Sil09, Proposition VII.5.5]). *There is a finite extension L of K such that E_L has good reduction if and only if $j(E) \in \mathcal{O}_K$.*

Problem 1.8 (Reading off the type).
Let $p \geqslant 5$, and let $\varepsilon \in \mathbb{Z}$ be a non-square modulo p. For each equation, decide whether it is minimal at p, and find the reduction type of the curve over $\mathbb{Q}_p$: good, split multiplicative, nonsplit multiplicative, or additive.

Solution. Here $a_4 = 0$, so (1.1) reads $\Delta = -16a_6(4a_2^3 + 27a_6)$ and $c_4 = 16a_2^2$. Since $p \geqslant 5$, the factor 16 is a unit.

(1) $y^2 = x^3 + x^2 + p$.

Solution. **Minimal. Split multiplicative.**

$$\Delta = -16p(27p + 4), \qquad v(\Delta) = 1 < 12, \qquad c_4 = 16,$$

so the equation is minimal (Proposition 1.3), and c_4 is a unit: multiplicative reduction (Proposition 1.5). The reduction $y^2 = x^3 + x^2 = x^2(x+1)$ has its node at $(0,0)$, where the lowest-degree terms $y^2 - x^2$ give the tangent lines $y = \pm x$. Their slopes lie in $\mathbb{F}_p$: split.

(2) $y^2 = x^3 + \varepsilon x^2 + p$.

Solution. **Minimal. Nonsplit multiplicative.**

$$\Delta = -16p(4\varepsilon^3 + 27p), \qquad v(\Delta) = 1, \qquad c_4 = 16\varepsilon^2,$$

using $p \nmid \varepsilon$. So c_4 is a unit: multiplicative. The tangent lines to $y^2 = x^2(x + \varepsilon)$ at $(0,0)$ are $y^2 = \varepsilon x^2$, with slopes $\pm\sqrt{\varepsilon} \notin \mathbb{F}_p$: nonsplit.

(3) $y^2 = x^3 + p$.

Solution. **Minimal. Additive.** $\Delta = -432p^2$ has $v(\Delta) = 2$, so the equation is minimal, and $c_4 = 0$: additive. The reduction $y^2 = x^3$ has a cusp at $(0,0)$.

(4) $y^2 = x^3 + p^6$.

Solution. **Not minimal. Good reduction.** Now $\Delta = -432p^{12}$ and $c_4 = 0$, so Proposition 1.3 does not apply, and the reduction $y^2 = x^3$ of this equation has a cusp. But the equation is not minimal: substituting $x = p^2x'$, $y = p^3y'$ and dividing by p^6 gives

$$y'^2 = x'^3 + 1, \qquad \Delta = -432,$$

a unit. So the curve has good reduction, and the cusp was an artifact of a bad model.

Remark. This is why reduction types are defined with a minimal equation.

(5) $y^2 = x^3 + x^2 + p^{12}$.

Solution. **Minimal. Split multiplicative.** $\Delta = -16p^{12}(27p^{12}+4)$ has $v(\Delta) = 12$, but $c_4 = 16$ is a unit, so the equation is minimal by Proposition 1.3: a scaling as in (4) would divide c_4 by u^4 and leave $\mathbb{Z}_p$. The reduction is the same as in (1).

Remark. Large $v(\Delta)$ alone does not make an equation non-minimal.

(6) Here is `Magma`'s opinion at $p = 7$ and $\varepsilon = 3$. For each curve it prints the valuation of the minimal discriminant, the Kodaira symbol of the reduction (I_0 for good reduction, I_n with $n \geqslant 1$ for multiplicative, anything else for additive), and the polynomial $1 - a_7T$, or $1 - a_7T + 7T^2$ in the case of good reduction.

```
R<T> := PolynomialRing(Integers());
p := 7; eps := 3;                          // 3 is not a square modulo 7
Info := function(a)
  E := EllipticCurve(a); L := LocalInformation(E, p);
  return <L[2], L[5], R!EulerFactor(E, p)>;
end function;
Info([0, 1, 0, 0, p]);                     // <1, I1, -T + 1>
Info([0, eps, 0, 0, p]);                   // <1, I1, T + 1>
Info([0, 0, 0, 0, p]);                     // <2, II, 1>
Info([0, 0, 0, 0, p^6]);                   // <0, I0, 7*T^2 + 4*T + 1>
Info([0, 1, 0, 0, p^12]);                  // <12, I12, -T + 1>
```

Check that it agrees with your answers. Note the values of a_7 at the curves with bad reduction. Problem 2.2 explains them.

Solution. **They agree.** I_1, I_1 and I_{12} are multiplicative, II is additive, and `Magma` finds the good reduction of (4), with $v(\Delta) = 0$ for its minimal equation:

	(1)	(2)	(3)	(4)	(5)
$v(\Delta)$, minimal	1	1	2	0	12
Kodaira symbol	I_1	I_1	II	I_0	I_{12}
reduction	split	nonsplit	additive	good	split
a_7	1	-1	0	-4	1

2. Counting rational points on the singular reduction

Definition 2.1 (The nonsingular locus)**.** When the equation of E is minimal, the nonsingular locus E_p^{ns} is E_p minus its singular point, if it has one.

Problem 2.2 (Counting with lines through the singular point)**.**
Let E be an elliptic curve over $\mathbb{Q}_p$ with bad reduction, and let S be the singular point of its reduction E_p.

(1) Show that $S \in E_p(\mathbb{F}_p)$, and that $S \neq O$.

Solution. By Proposition 1.5 and Proposition 1.4, E_p has exactly one singular point S over $\overline{\mathbb{F}}_p$. The Frobenius automorphism σ of $\overline{\mathbb{F}}_p$ maps singular points of E_p to singular points, because E_p and its partial derivatives have coefficients in $\mathbb{F}_p$. So $\sigma(S) = S$, and $S \in E_p(\mathbb{F}_p)$. The rational point $O = [0:1:0]$ is nonsingular: in the chart $Y = 1$ the equation is

$$z + a_1xz + a_3z^2 = x^3 + a_2x^2z + a_4xz^2 + a_6z^3,$$

whose linear part z does not vanish.

(2) Show that a change of variables over $\mathbb{F}_p$ moves S to $(0,0)$ and puts E_p in the form

$$y^2 + a_1xy = x^3 + a_2x^2,$$

and that the tangent lines at S are the lines $y = \alpha x$ with $g(\alpha) = 0$, where $g(t) \coloneqq t^2 + a_1t - a_2$.

Solution. Write $S = (x_0, y_0)$ and substitute $x + x_0$ for x and $y + y_0$ for y. This keeps the Weierstrass form and has coefficients in $\mathbb{F}_p$. In the new equation

$$f(x,y) = y^2 + a_1xy + a_3y - x^3 - a_2x^2 - a_4x - a_6 = 0,$$

the rational point $(0,0)$ lies on the curve, so $a_6 = 0$, and is singular, so

$$a_4 = -\frac{\partial f}{\partial x}(0,0) = 0, \qquad a_3 = \frac{\partial f}{\partial y}(0,0) = 0.$$

The tangent lines at $(0,0)$ are given by the terms of lowest degree, $y^2 + a_1xy - a_2x^2$, which factor as $(y - \alpha x)(y - \alpha' x)$ with α, α' the roots of g.

(3) Let $t \in \mathbb{F}_p$. Show that the line $y = tx$ meets E_p at S and, if $g(t) \neq 0$, at exactly one more $\mathbb{F}_p$-rational point $P_t = \big(g(t), t\,g(t)\big)$, and nowhere else in the affine plane. Show that $t \mapsto P_t$ is a bijection from $\{t \in \mathbb{F}_p : g(t) \neq 0\}$ onto $E_p^{\mathrm{ns}}(\mathbb{F}_p) - \{O\}$.

Solution. Substituting $y = tx$ gives

$$t^2x^2 + a_1tx^2 = x^3 + a_2x^2, \quad \text{that is,} \quad x^2\big(g(t) - x\big) = 0.$$

So the affine points of E_p on the line are S and, if $g(t) \neq 0$, the point P_t. If $g(t) = 0$, the line is tangent at S and meets E_p only there. P_t is nonsingular, since $P_t \neq S$. Conversely, a nonsingular affine point (x,y) has $x \neq 0$, since $x = 0$ forces $y^2 = 0$, which is S. So it lies on the line of slope $t = y/x$ through S, with $g(t) \neq 0$, and equals P_t. The two maps are inverse to each other.

(4) Deduce that $\#E_p^{\mathrm{ns}}(\mathbb{F}_p) = p + 1 - r$, where r is the number of roots of g in $\mathbb{F}_p$, and hence that

$$a_p = \begin{cases} 1 & \text{for split multiplicative reduction,} \\ -1 & \text{for nonsplit multiplicative reduction,} \\ 0 & \text{for additive reduction.} \end{cases}$$

Compare with the output in Problem 1.8(6).

Solution. By (3), $\#E_p^{\mathrm{ns}}(\mathbb{F}_p) = 1 + (p - r)$, counting O. Adding S,

$$\#E_p(\mathbb{F}_p) = p + 2 - r, \qquad \text{so} \qquad a_p = r - 1.$$

By Definition 1.1 and (2), $r = 2$ for split multiplicative reduction, $r = 0$ for nonsplit, and $r = 1$ for additive reduction, where g is a square. So $a_p = 1, -1, 0$. This matches Problem 1.8(6).

Remark. The three values are the numbers of $\mathbb{F}_p$-points of $\mathbb{G}_{\mathrm{m}}$, of a twisted multiplicative group and of $\mathbb{G}_{\mathrm{a}}$: E_p^{ns} is isomorphic to one of these as a group [Sil09, Proposition III.2.5], [Poo08, §6.3].

(5) Check (4) by hand for $y^2 = x^3 + x^2$ and $y^2 = x^3 + 2x^2$ over $\mathbb{F}_5$.

Solution. **$a_5 = 1$ and $a_5 = -1$.** For $y^2 = x^3 + x^2$ over $\mathbb{F}_5$, $g(t) = t^2 - 1$ has the roots ± 1, and $P_t = (t^2 - 1, t^3 - t)$ for $t = 0, 2, 3$ gives $(4,0)$, $(3,1)$ and $(3,4)$. With O and S there are 5 rational points, so $a_5 = 1$. For $y^2 = x^3 + 2x^2$, $g(t) = t^2 - 2$ has no roots, since 2 is not a square modulo 5, and $t = 0, \ldots, 4$ gives $(3,0)$, $(4,4)$, $(2,4)$, $(2,1)$ and $(4,1)$: with O and S, 7 rational points, so $a_5 = -1$.

3. Changing the curve, changing the field

Problem 3.1 (Twists and extensions)**.**
Let $p \geqslant 5$. For $d \in \mathbb{Q}_p{}^\times$ and $E\colon y^2 = x^3 + a_2x^2 + a_4x + a_6$, the quadratic twist of E by d is

$$E^d\colon y^2 = x^3 + da_2x^2 + d^2a_4x + d^3a_6.$$

(1) Show that $(x, y) \mapsto (dx, d^{3/2}y)$ is an isomorphism $E \to E^d$ over $\mathbb{Q}_p(\sqrt{d})$, and that $\Delta(E^d) = d^6\Delta(E)$, $c_4(E^d) = d^2c_4(E)$ and $j(E^d) = j(E)$.

Solution. If $y^2 = x^3 + a_2x^2 + a_4x + a_6$, then

$$(d^{3/2}y)^2 = d^3y^2 = (dx)^3 + da_2(dx)^2 + d^2a_4(dx) + d^3a_6,$$

so the image lies on E^d, and $(x, y) \mapsto (x/d, d^{-3/2}y)$ is the inverse. This is a change of variables with $u = d^{-1/2}$, which multiplies Δ by $u^{-12} = d^6$ and c_4 by $u^{-4} = d^2$ [Sil09, Table 3.1]. Hence $j = c_4^3/\Delta$ is unchanged.

Remark. Or use (1.1): every term of Δ is multiplied by d^6, and every term of c_4 by d^2.

From now on, $E\colon y^2 = x^3 + x^2 + p$ is the curve of Problem 1.8(1), with split multiplicative reduction.

(2) Let $\varepsilon \in \mathbb{Z}$ be a non-square modulo p. Show that E^ε has nonsplit multiplicative reduction, and that $a_p(E^\varepsilon) = -a_p(E)$. Since $\mathbb{Q}_p(\sqrt{\varepsilon})$ is an unramified extension of $\mathbb{Q}_p$, what does this say about which features of the reduction are preserved by unramified extensions?

Solution. **Unramified extensions preserve the reduction type, but not split versus nonsplit.** $E^\varepsilon\colon y^2 = x^3 + \varepsilon x^2 + \varepsilon^3 p$ has $v(\Delta) = v(\Delta(E)) = 1$ and $v(c_4) = 0$, so it is minimal with multiplicative reduction, and its reduction $y^2 = x^2(x + \varepsilon)$ has tangent slopes $\pm\sqrt{\varepsilon} \notin \mathbb{F}_p$: nonsplit. By Problem 2.2, $a_p(E) = 1$ and $a_p(E^\varepsilon) = -1$. Over the unramified extension $\mathbb{Q}_p(\sqrt{\varepsilon})$, whose residue field is $\mathbb{F}_{p^2}$, the two curves become isomorphic, and both have split multiplicative reduction there.

(3) Show that $E^p\colon y^2 = x^3 + px^2 + p^4$ has additive reduction. Over which quadratic extension of $\mathbb{Q}_p$ does it acquire multiplicative reduction? Show that it has good reduction over no finite extension of $\mathbb{Q}_p$.

Solution. **Additive. Multiplicative over $\mathbb{Q}_p(\sqrt{p})$. Never good.** By (1),

$$v(\Delta(E^p)) = 6 + 1 = 7 < 12, \qquad v(c_4(E^p)) = v(16p^2) = 2 > 0,$$

so the equation is minimal, with additive reduction. Its reduction is $y^2 = x^3$. Over $\mathbb{Q}_p(\sqrt{p})$, $E^p \cong E$ by (1), and E keeps its multiplicative reduction by Proposition 1.6. Finally $v(j) = 3v(c_4) - v(\Delta) = -1$ for E, and hence for E^p, so $j(E^p) \notin \mathbb{Z}_p$ and E^p has good reduction over no finite extension (Proposition 1.7).

(4) Let $E_3\colon y^2 = x^3 + p$, and let L be a finite extension of $\mathbb{Q}_p$ with ramification index e. Show that E_3 has good reduction over L if and only if $6 \mid e$.

Solution. Over L, $v_L(\Delta) = e\,v(-432p^2) = 2e$. If E_3 has good reduction over L, a minimal equation over L has $v_L(\Delta') = 0$, and $\Delta' = u^{-12}\Delta$ by Proposition 1.3, so $2e = 12v_L(u)$ and $6 \mid e$. Conversely, if $6 \mid e$, write $p = \varpi^e w$ and substitute $x = \varpi^{e/3}x'$, $y = \varpi^{e/2}y'$. Dividing by ϖ^e gives

$$y'^2 = x'^3 + w, \qquad \Delta = -432w^2,$$

a unit: good reduction. For instance $L = \mathbb{Q}_p(p^{1/6})$ works.

(5) Show that $y^2 = x^3 + px$ has additive reduction, and good reduction over L as in (4) if and only if $4 \mid e$.

Solution. **Additive. Good over L if and only if $4 \mid e$.** Here $\Delta = -64p^3$ and $c_4 = -48p$, so $v(\Delta) = 3$ and $v(c_4) = 1$: minimal, with additive reduction. Over L, $v_L(\Delta) = 3e$, and the argument of (4) gives good reduction if and only if $12 \mid 3e$, that is, $4 \mid e$. For "if", substitute $x = \varpi^{e/2}x'$, $y = \varpi^{3e/4}y'$ to get

$$y'^2 = x'^3 + wx', \qquad \Delta = -64w^3,$$

a unit.

Remark. Both E_3 and this curve have $j \in \mathbb{Z}_p$ (0 and 1728), as Proposition 1.7 requires.

4. ⋆ Frey curves

This section is [Sil09, Exercise 8.23], with the conductor of [Mil20, Aside II.3.5]. The LMFDB [LMF25] is the answer key.

Problem 4.1 (Frey curves are semistable)**.**
An elliptic curve over $\mathbb{Q}$ is semistable if it has good or multiplicative reduction at every prime. Let A, B, C be nonzero integers with $A + B = C$ and $\gcd(A, B, C) = 1$. Let

$$E\colon y^2 = x(x + A)(x - B).$$

(1) Show that $\Delta = 16A^2B^2C^2$ and $c_4 = 16(A^2 + AB + B^2)$.

Solution. The roots of $x(x+A)(x-B)$ are 0, $-A$ and B, with differences A, B and $A+B = C$. For $y^2 = (x - e_1)(x - e_2)(x - e_3)$,

$$\Delta = 16\prod_{i<j}(e_i - e_j)^2, \qquad c_4 = 16\big(e_1^2 + e_2^2 + e_3^2 - e_1e_2 - e_1e_3 - e_2e_3\big),$$

which follow from (1.1) with $a_2 = -(e_1 + e_2 + e_3)$, $a_4 = e_1e_2 + e_1e_3 + e_2e_3$ and $a_6 = -e_1e_2e_3$. Here they give $\Delta = 16A^2B^2C^2$ and $c_4 = 16(A^2 + B^2 + AB)$.

(2) Let p be an odd prime dividing ABC. Show that the equation is minimal at p and that E has multiplicative reduction at p. Show that the reduction is split if and only if $-B$, A or B is a square modulo p, according as p divides A, B or C.

Solution. Since A, B, C are pairwise coprime ($A + B = C$ and $\gcd(A, B, C) = 1$), p divides exactly one of them. If $p \mid A$, then $c_4 \equiv 16B^2$. If $p \mid B$, then $c_4 \equiv 16A^2$. If $p \mid C$, then $B \equiv -A$ and $c_4 \equiv 16A^2$. In each case c_4 is a unit, so the equation is minimal (Proposition 1.3) and the reduction is multiplicative (Proposition 1.5). The reduction and the tangent lines at its node are:

- $p \mid A$: $y^2 = x^2(x - B)$, node $(0, 0)$, tangents $y^2 = -Bx^2$;
- $p \mid B$: $y^2 = x^2(x + A)$, node $(0, 0)$, tangents $y^2 = Ax^2$;
- $p \mid C$: $y^2 = x(x + A)^2$, node $(-A, 0)$, tangents $y^2 = -A(x + A)^2$, and $-A \equiv B$.

So the slopes lie in $\mathbb{F}_p$ exactly when $-B$, A, respectively B, is a square modulo p.

(3) Suppose that $A \equiv 0 \pmod{16}$ and $B \equiv 3 \pmod 4$. Show that the substitution $x = 4x'$, $y = 8y' + 4x'$ gives an equation with coefficients in $\mathbb{Z}$,

$$y'^2 + x'y' = x'^3 + \frac{A - B - 1}{4}\,x'^2 - \frac{AB}{16}\,x',$$

with discriminant $(ABC)^2/2^8$ and $c_4 = A^2 + AB + B^2$. Deduce that E has good reduction at 2 if $v_2(A) = 4$, and multiplicative reduction at 2 if $v_2(A) \geqslant 5$. So E is semistable.

SOLUTION. Substituting and dividing by 64, $(8y' + 4x')^2 = 4x'(4x' + A)(4x' - B)$ becomes the displayed equation of the statement. The coefficients are integers because $A - B - 1 \equiv 0 - 3 - 1 \equiv 0 \pmod 4$ and $16 \mid A$. This is a change of variables with $u = 2$, so (Proposition 1.3)

$$\Delta' = 2^{-12}\Delta = (ABC)^2/2^8, \qquad c_4' = 2^{-4}c_4 = A^2 + AB + B^2.$$

Since A is even and B odd, c_4' is odd, so the new equation is minimal at 2. Since B and C are odd, $v_2(\Delta') = 2v_2(A) - 8$: good reduction if $v_2(A) = 4$, and multiplicative reduction if $v_2(A) \geqslant 5$ (Proposition 1.5). With (2), and good reduction at the odd primes not dividing ABC, E is semistable. (At odd p, the substitution has $u = 2$, a unit, so it does not change the reduction type there.)

(4) The conductor of E is $N = \prod_p p^{f_p}$, where $f_p = 0$ if E has good reduction at p, $f_p = 1$ if it has multiplicative reduction, and $f_p \geqslant 2$ if it has additive reduction. Under the hypotheses of part (3), compute N in terms of A, B, C. Find the curves for $(A, B, C) = (16, -1, 15)$ and $(32, -5, 27)$ in the LMFDB, and check their conductors and their split and nonsplit primes against your answers.

SOLUTION. ***N* is the product of the odd primes dividing *ABC*, times 2 if 32 | *A*.** Every f_p is 0 or 1, and $f_p = 1$ exactly at the odd primes dividing ABC, and at 2 when $32 \mid A$. For $(16, -1, 15)$, $N = 15$: the curve is 15a3 in Cremona's labels, with split multiplicative reduction at 5 ($B = -1$ is a square modulo 5) and nonsplit at 3 (-1 is not a square modulo 3). For $(32, -5, 27)$, $N = 30$: the curve is 30a2, split at 3 ($B = -5 \equiv 1$), nonsplit at 5 ($A = 32 \equiv 2$ is not a square modulo 5), and multiplicative at 2.

(5) Suppose that $a^\ell + b^\ell = c^\ell$ with $\ell \geqslant 5$ prime and a, b, c nonzero coprime integers. Show that, after reordering and changing signs, $(A, B, C) = (a^\ell, b^\ell, c^\ell)$ satisfies the hypotheses of part (3), and that the resulting curve has conductor $\prod_{p \mid abc} p$ and minimal discriminant $(abc)^{2\ell}/2^8$.

SOLUTION. Exactly one of a, b, c is even, since they are coprime and $a^\ell + b^\ell = c^\ell$. Since ℓ is odd, $x \mapsto x^\ell$ commutes with signs, so the relation can be rewritten as $A + B = C$ with A the ℓ-th power of the even one: $a^\ell + b^\ell = c^\ell$ itself if a is even, and $c^\ell + (-b)^\ell = a^\ell$ if c is even. Then $v_2(A) \geqslant \ell \geqslant 5$. If $B \equiv 1 \pmod 4$, replace (A, B, C) by $(-A, -B, -C)$. By (4) and (3), the conductor is the product of all primes dividing ABC, which is $\prod_{p \mid abc} p$, and the minimal discriminant is $(ABC)^2/2^8 = (abc)^{2\ell}/2^8$, an ℓ-th power up to 2^8.

Remark. Frey observed that such a curve would be very strange. Ribet proved that, if it were modular, its ℓ-torsion would come from a modular form of level 2, and there is none. Wiles and Taylor–Wiles proved that semistable elliptic curves over $\mathbb{Q}$ are modular, so the curve, and the solution, cannot exist [Mil20, §V.9].

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